

Volume 6 — Reference, Glossary & Index

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

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Chapter 1 — Original Mazak Manual Library

All originals live in `OneDrive-Personal/OneDrive Docs/Mazak/Manuals_SN060231/` (synced locally and shared via OneDrive). Source: Mazak Corporation secure download, 2025-10-13.

Publication	Title	Folder	Retrofit relevance
413S038	Operating Manual — VQC-20/40 & 20/50	01_Operating	Machine handling basics; Mazatrol operation now historical
m00s075	Operating Manual — CAM M-2	01_Operating	Historical (Mazatrol UI retired)
413m033	Maintenance Manual — VQC-20/40 & 20/50	02_Maintenance	Active authority — mechanics, lube, hydraulics
41434WB	Electrical Circuit Diagram (M-2), 6/1984	02_Maintenance	Active authority — machine-side wiring; primary retrofit source
YM2V39L	PLC Ladder Diagrams	02_Maintenance	Active — interlock behavior to reproduce in HAL (385-signal element list extracted)
PAREXM210E	Parameter List & Explanation — M-2	03_Parameters	Historical + useful for decoding legacy setup values
M00P111	Programming Manual — CAM M-2 (EIA/ISO)	04_Supplements	G-code reference for legacy programs
m00p135	Programming Manual — M-2E	04_Supplements	Reference only — this machine is M-2, not M-2E
M00P018	Programming Manual — Mazatrol CAM 3D	04_Supplements	Option — verify if installed on SN 060231
M00T002	Tooling Manual — Mazak Machining Center	04_Supplements	Active — holders and pull studs

413LE02A000	Parts List — VQC 20/40 & 50	05_Parts	Active — spare parts sourcing
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Also in that folder: `servo_amp_analysis.pdf/.md` , `servo_topology_p082.png` , `terminal_unit_detail_p085.png` , cabinet photos (`photos_2026-07/`).

Chapter 2 — Key Project Files

File	Path	Role
Architecture decision	GitHub/Github LinuxCNC/docs/architecture_decision.md	Why 7i97T + 7i49 + 7i84U
Project status / TODO	GitHub/Github LinuxCNC/docs/project_status.md	Live checklist and bring-up order
Pin authority	GitHub/Github LinuxCNC/mesa/current_pin_authority.csv	The wiring truth table (supersedes signal_map.csv)
Wiring master workbook	GitHub/Github LinuxCNC/wiring/Mazak_Wiring_Master_7i97T_7i49_7i84U.xlsx	Consolidated workbook; legacy tabs clearly separated
I/O workbook	GitHub/Github LinuxCNC/bom/Mazak_VQC_20-40_Retrofit_IO_Workbook.xlsx	Original I/O planning source
HAL/INI skeletons	GitHub/Github LinuxCNC/linuxcnc/	Bring-up configs (placeholders until pin dump)
Firmware checklist	GitHub/Github LinuxCNC/mesa/mesa_firmware_checklist.md	Facts to collect before locking HAL names
PLC element list	Manuals_SN060231/02_Maintenance/YM2V39L_element_list.csv	385 legacy signals: 179 X-inputs, 206 Y-outputs

Chapter 3 — External References

- Mesa 7i49 manual (excitation, W2 jumper, RESDRV/RESSIN/RESCOS): mesanet.com/pdf/motion/7i49man.pdf
- Sister retrofit — `srcdco/MazakVQC1540` (VQC 15/40, same TRA drives/HD81-12S motors, plain 7i49 @ 5 kHz), incl. the complete 2017-05-01 reference package
- LinuxCNC forum: servo PID tuning thread (VQC 15/40) and the owner's build thread "Mesa conversion for a Mazak VQC 20/40 M2 mill"
- Mitsubishi TRA-31 drive manual; Meldas YM2 / Mazatrol M2 maintenance manual (Mitsubishi FA knowledge base)

Chapter 4 — Glossary

Term	Meaning
ATC	Automatic Tool Changer — hydraulic arm + magazine on this machine
BCD	Binary-Coded Decimal — how magazine pocket position is reported (PRS-21...25)
Bring-up	Staged first-power procedure proving each subsystem safely (Vol 3 §8)
ENA / S-ON	Drive enable signal from 7i97T TB3 to a TRA servo drive
E-stop chain	Series-wired hardware circuit that drops contactors directly
FR-SX	Mitsubishi spindle drive taking an analog speed reference
HAL	Hardware Abstraction Layer — LinuxCNC's signal-wiring layer
hm2_eth	LinuxCNC driver for Ethernet-connected Mesa FPGA boards
INI	LinuxCNC machine configuration file
Interposing relay	Small relay (RLY-1...7) between a Mesa output and a heavy/100 VAC coil
Mazatrol M-2	Original conversational CNC control (retired by this retrofit)
mesaflash	Utility to query/flash Mesa boards (<code>--readhmid</code>)
PLAN_VERIFY	Pin-authority status: engineered plan, must be traced before wiring
PRS / PHS	Proximity / photo-sensor switch designators from the original ladder
Resolver	Analog rotary position transducer; absolute within one rev; read by the 7i49
RESDRV / RESSIN / RESCOS	7i49 excitation drive and sin/cos return terminals
RLY	Retrofit-added interposing relay
Smart-serial (sserial)	Mesa's serial field-bus linking 7i97T TB4 to the 7i84U
SOL	Hydraulic/pneumatic solenoid valve designator
SSR	Solid-state relay output (7i97T TB5.13–24)
Tachometer	Analog speed sensor closing the TRA drive's velocity loop
TRA	Mitsubishi analog servo drive family retained in this machine
W2 jumper	7i49 half-excitation option — field-verification use only
WHB04B	Wireless USB MPG pendant (optional)

Chapter 5 — Alphabetical Index

References: **Volume** · chapter/§. Example: V4 §3 = Volume 4, section 3.

A

- Air blast (SOL-62): V2 c8; V4 §4
- Alarm output: V2 c8; V4 §9
- Alarms, operator response: V2 c12
- Analog command wiring: V3 c8; V4 §2
- Analog polarity / runaway hazard: V1 c6; V3 c8; V5 c6
- Architecture (selected): V3 c1; V4 §1
- Architecture (rejected 6i25/7i77): V3 c10
- ATC operation: V2 c6 · troubleshooting: V5 c8 · wiring: V4 §§8–9

B

- BCD magazine coding: V2 c6; V4 §8; V5 c8
- Bring-up procedure: V3 c8
- Brake, Z axis (N1J-L2-201): V4 §9

C

- Cabinet photo checklist: V3 c9
- Coolant control & level: V2 c8; V4 §§8–9
- Cycle start / feed hold / single block: V2 c9; V4 §8

D

- Daily maintenance: V5 c2
- Door interlock: V1 c4; V4 §8
- Drive enables (S-ON): V4 §2; V5 c6
- Drive faults: V2 c12; V4 §8; V5 c4

E

- E-stop chain: V1 c4; V4 §7 · functional test: V3 c9
- Ethernet setup (hm2_eth, 192.168.1.121): V3 c3
- Excitation, resolver (5 kHz): V3 c5; V4 §3

F–G

- Field power banks (7i84U): V3 c6; V4 §8
- FR-SX spindle drive: V2 c5; V4 §6; V5 c7
- G-code programs: V2 c7
- Gear shift Hi/Lo: V2 c5; V4 §§6, 8–9
- Glossary: V6 c4

H

- HAL files: V3 c7
- Homing: V2 c3; V5 c6
- HR-11F-24 24 V supply: V1 c1; V3 c8
- Hydraulic pressure switch (Sanwa): V4 §8; V5 c3

I–L

- Index of original manuals: V6 c1
- Interposing relays RLY-1...7: V1 c6; V4 §§4, 9
- Latency test: V3 c3

- Limits (overtravel): V2 c12; V4 §4
- Lockout/tagout: V1 c5
- Lubrication: V5 c3

M

- Magazine: V2 c6; V4 §8; V5 c8
- Maintenance schedules: V5 c2
- Mazatrol M-2 (history): V1 c1–2
- Mesa 7i49: V3 c5; V4 §3
- Mesa 7i84U: V3 c6; V4 §§5, 8–9
- Mesa 7i97T: V3 c4; V4 §2, 4–5
- mesaflash / readhmid: V3 c3; V5 c9

O–P

- Orient, spindle: V2 c5; V5 c7
- Pendant (WHB04B): V2 c10
- Pin authority table & status codes: V4 §10; V6 c2
- PLC ladders (YM2V39L): V1 c2; V5 c8; V6 c1
- Power-up / shutdown: V2 c2, c11

R

- Resolver wiring & warnings: V3 c5; V4 §3
- Resolver troubleshooting: V5 c4, c6
- Runaway (axis): V1 c6; V5 c6

S

- Safety rules: V1 c3
- Smart-serial link: V4 §5; V5 c9
- Spare parts: V5 c10
- Spindle operation: V2 c5 · wiring: V4 §6 · troubleshooting: V5 c7
- Spindle enable conflict (TB3 vs 7i84U): V3 c8; V4 §2, 6, 9
- SSR outputs: V4 §4

T

- Tamagawa resolvers: V1 c1; V3 c5; V4 §3
- TB3 / TB4 / TB5 maps: V4 §§2, 5, 4
- Tool changes: V2 c6
- Tool table (tool.tbl): V2 c6–7; V5 c2
- Tooling (M00T002): V1 c3; V6 c1
- TRA servo drives: V1 c1; V4 §2; V5 c4
- Transformer, 460 V: V1 c1, c5

V–Z

- Verification checklists: V3 c9
 - W2 half-drive jumper: V3 c5; V4 §3
 - Wiring diagrams: V4 (all)
 - YM2V39L element list: V6 c1–2
 - Z brake release: V4 §9
 - 24 VDC distribution: V3 c8; V4 §§4, 8
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End of Volume 6 — end of manual set. Revision A, compiled July 2026 from project sources listed in Volume 0.